

Swiggy Data Warehouse Analytics Using MySQL

Building a Star Schema Data Warehouse and Generating Business Insights from
197,403 Food Delivery Orders



Executive Summary

197K

Total Orders

Analyzed across all cities and restaurants

₹53M

Total Revenue

Captured in the data warehouse

4.34

Avg Rating

Across all restaurants and categories

5

Dimension Tables

Star Schema with 1 central fact table

Objective

Transform raw food delivery data into an analytics-ready warehouse using MySQL, enabling SQL-driven business insights for growth strategy.

Deliverable

A fully optimized **Star Schema** with **fact_swiggy_orders** at the center, surrounded by 5 dimension tables — powering actionable reporting across geography, time, restaurant, category, and dish.

Project Workflow



Data Cleaning

Null validation, date conversion, duplicate removal, and consistency checks

Data Modeling

Star Schema design with **fact_swiggy_orders** + 5 dimension tables

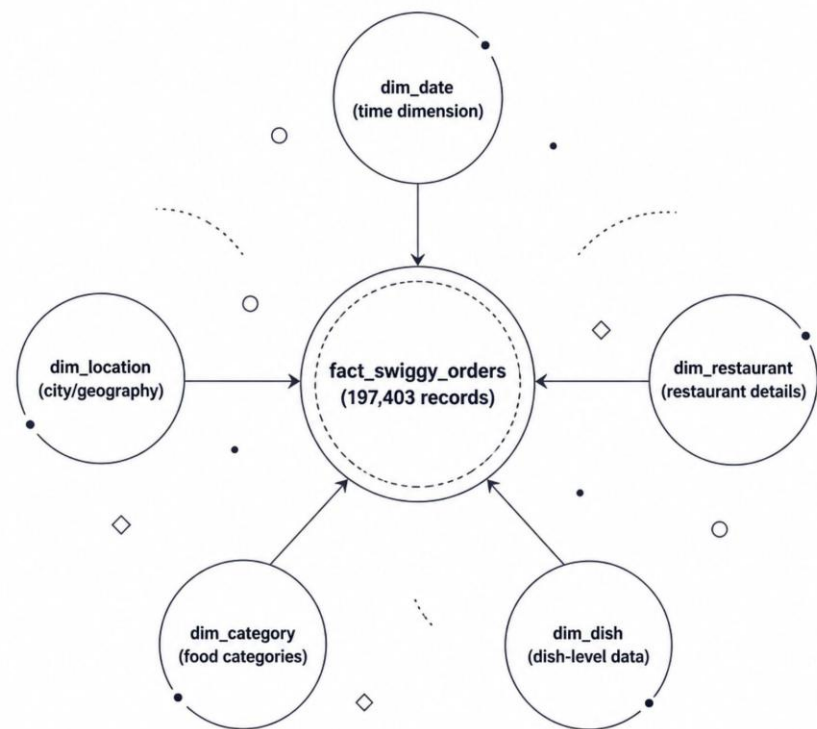
ETL Process

Dimension loading, fact table loading, and query optimization

Analytics

SQL-driven insights for business decision-making

Star Schema Architecture



Fact Table

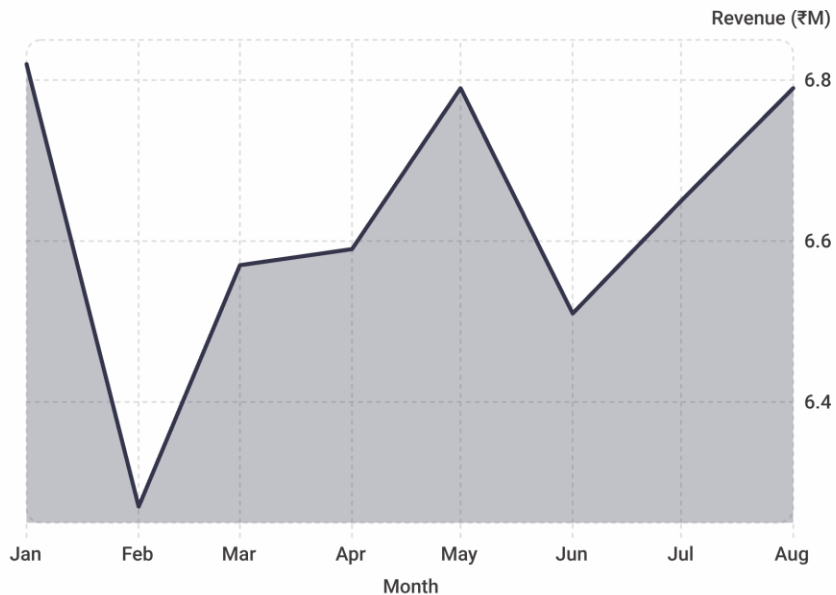
fact_swiggy_orders – 197,403 records capturing every transaction with foreign keys to all dimensions.

Dimension Tables

Indexed joins and optimized queries ensure fast reporting across all dimensions.

dim_date Day, month, year, weekday	dim_location City and geographic data	dim_restaurant Brand, type, rating	dim_category Food category classification	dim_dish Dish name and price
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Revenue Performance



Key Metrics

Insight

Revenue remained **remarkably stable** throughout the year, with **January generating the highest revenue** at ₹6.82M. The consistency signals a mature, loyal customer base with predictable ordering behavior — a strong foundation for forecasting and inventory planning.

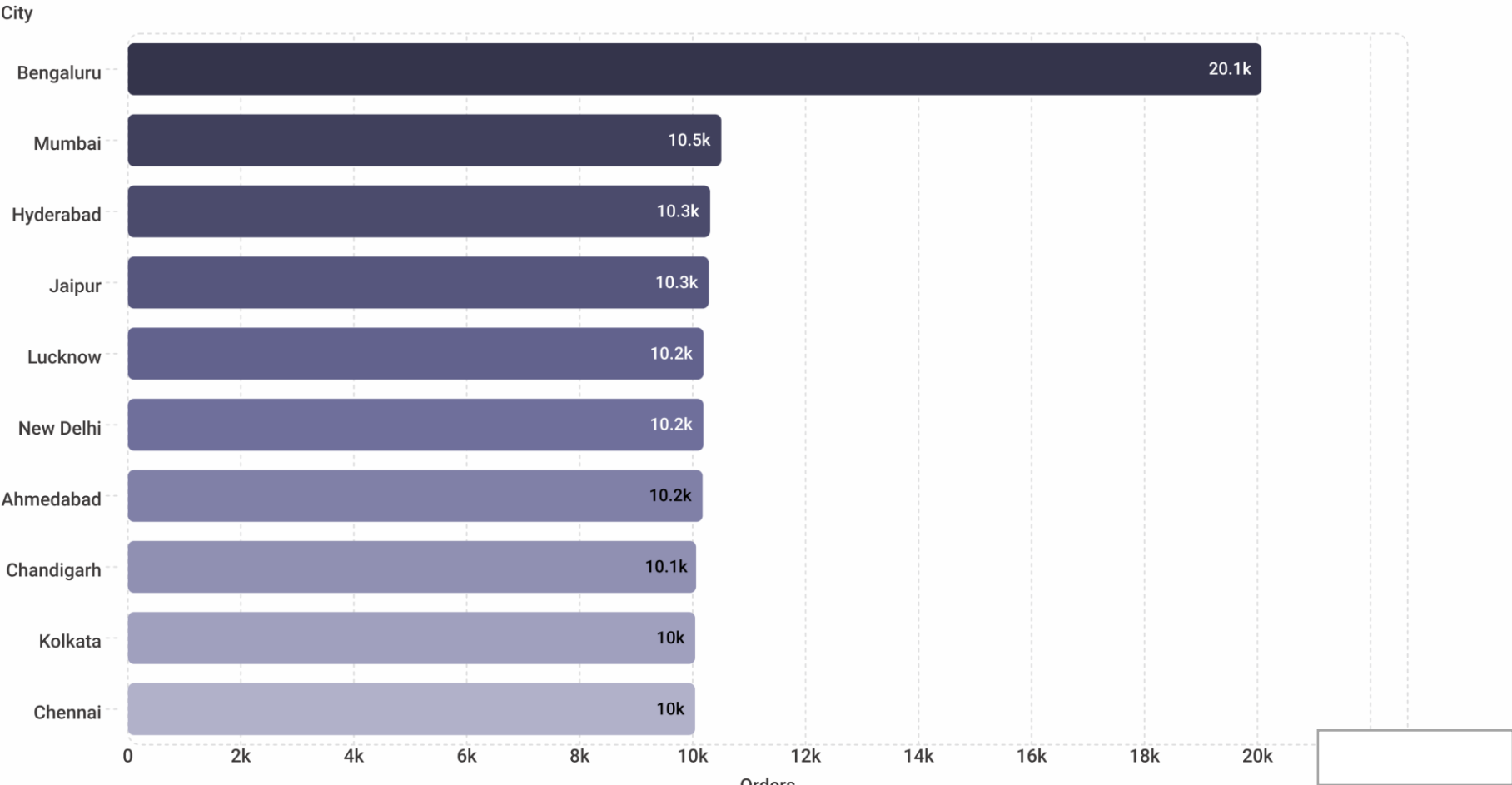
₹53 Million

Total Revenue

₹268.50

Avg Dish Price

Geographic Performance: Top Cities

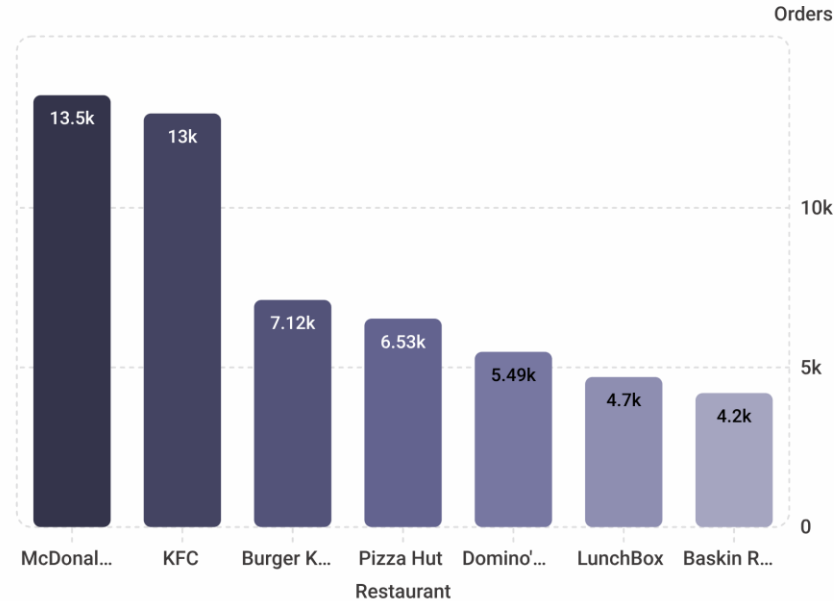


Restaurant Performance: QSR Dominance

What the Data Shows

Quick Service Restaurant (QSR) brands overwhelmingly dominate customer demand on the Swiggy platform. McDonald's leads with 13,528 orders, followed closely by KFC at 12,957 — together accounting for a disproportionate share of total volume.

- 1 McDonald's — 13,528 orders
- 2 KFC — 12,957 orders
- 3 Burger King — 7,115 orders
- 4 Pizza Hut — 6,529 orders
- 5 Domino's Pizza — 5,489 orders



Category Performance & Ratings

Top by Volume

Recommended

24,098 orders – highest volume category

Highest Rated Categories

Namkeen

★ 4.51 rating

Sweets

★ 4.46 rating

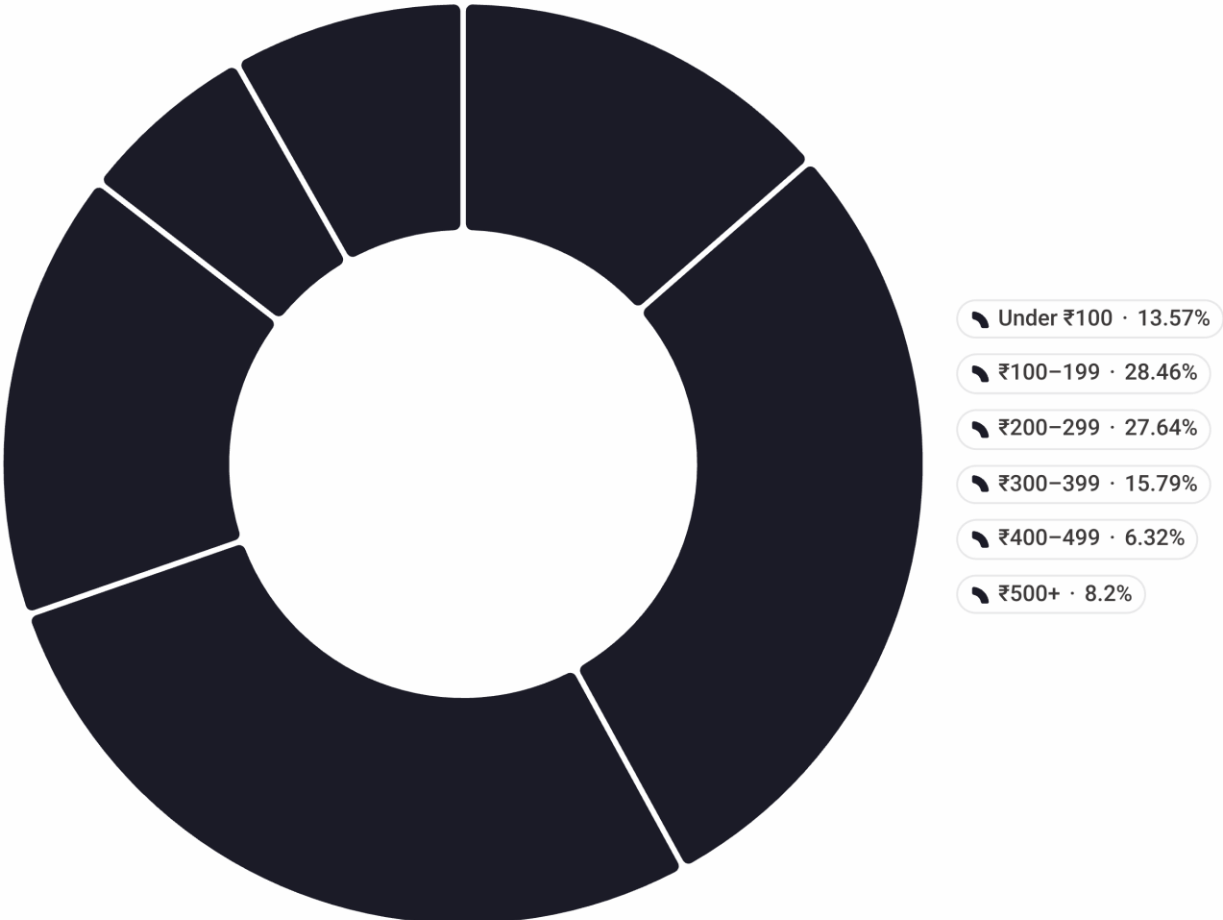
McSaver Combos

★ 4.41 rating

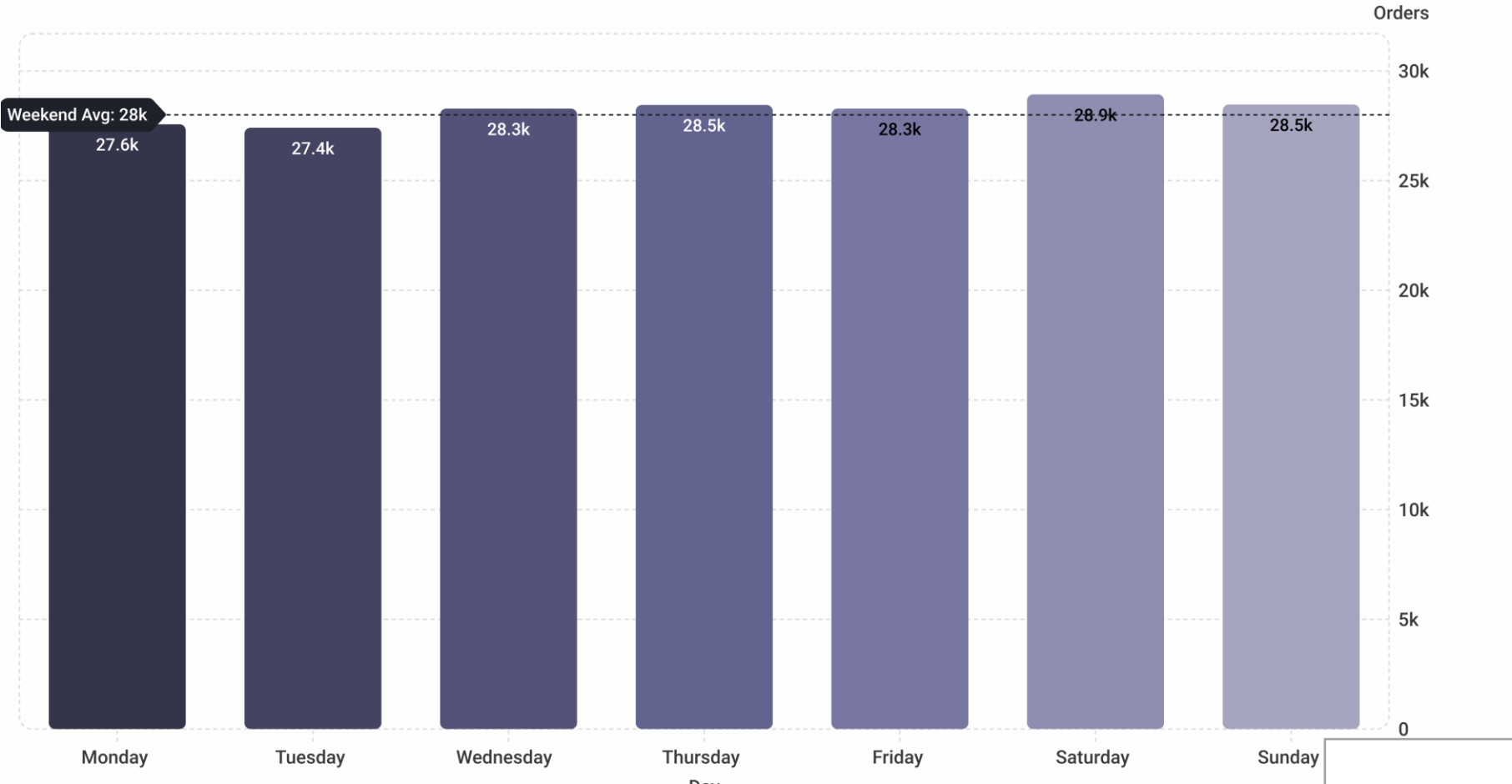
📌 High-rated categories achieve premium customer satisfaction despite lower order volume – a signal for targeted promotion and upsell opportunities.



Customer Spending Segmentation



Day-wise Order Patterns



Recommendations, Optimization & Business Impact

Business Recommendations

- Double down on Bengaluru
Increase marketing investments — 2x order volume vs. all other cities
- Target the ₹100–₹299 segment
56% of all orders — highest ROI for promotional offers
- Expand weekend delivery resources
3% higher demand on weekends; Saturday is peak
- Promote high-rated categories
Namkeen, Sweets, and McSaver Combos drive premium satisfaction
- Strengthen QSR partnerships
McDonald's and KFC lead demand — deepen co-marketing ties

Technical Optimization

Indexing Strategy

Optimized key columns for fast retrieval across all dimension joins

Query Optimization

Efficient joins between fact and dimension tables for reporting speed

Business Impact

₹53M

Scalable warehouse built for this transaction volume

7

Actionable insights generated from 197,403 orders



Enabled data-driven decision-making for Swiggy's growth strategy through optimized SQL reporting.

Full SQL Scripts & Documentation:

Link : github.com/harsh19122001/Swiggy-Data-Warehouse-Analytics-MySQL

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